

IoT-Enabled Fire Fighting Robot

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Declaration

We, hereby, declare that the work presented in this Project report is the outcome of the investigation performed by us under the supervision of Nazmun Nahid, Assistant Professor, Department of Computer Science, University of Asia Pacific. We also declare that no part of this Project report and thereof has been or is being submitted elsewhere for the award of any degree or Diploma..

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Abstract

Fire accidents are among the most devastating hazards in developing countries like Bangladesh, causing significant loss of life and property each year. This report presents the design, implementation, and testing of an IoT-based firefighting robot capable of remote operation, real-time environmental monitoring, and manual fire suppression from a safe distance. The system is built around a Wemos D1 R32 microcontroller paired with a Sensor Shield v5.0, an ESP32-CAM module for live video streaming, an L298N motor driver, four DC motors, flame sensors, an MQ2 gas sensor, a DHT11 temperature and humidity sensor, a buzzer, a 4-channel relay module, an SG90 servo motor, and a mini water pump. Control is delivered over Wi-Fi through a mobile web interface, where a human operator issues directional commands (forward, backward, left, right) and spray commands (spray on, spray off) while monitoring the live camera feed and sensor dashboard. Testing in a controlled environment confirmed reliable motor response, accurate fire and smoke detection, successful pump activation, and stable video streaming at approximately 15 fps with command latency below 1.5 seconds. The system demonstrates an effective, affordable proof-of-concept for reducing human exposure to fire hazards in small-to-medium scale incidents.

Contents

List of Figures	vi
List of Tables	vii
1 Introduction	1
1.1 Background and Motivation	1
1.2 Problem Statement	1
1.3 Objectives of the Project	2
1.4 Scope of the Work	2
1.5 Expected Outcomes	2
1.6 Impacts of the Project	3
1.6.1 Impact on Societal and Cultural Issues	3
1.6.2 Impact on Health, Safety, and Legal Issues	4
1.6.3 Impact on Environment and Sustainability Issues	4
2 Background and Literature Review	5
2.1 Preliminaries	5
2.2 Related Works / Existing Systems	7

2.3	Comparative Analysis of Related Works	7
2.4	Research Gap / Limitations of Existing Methods	8
2.5	Summary	8
3	System Analysis and Design	9
3.1	Requirement Analysis	9
3.2	System Overview / Architecture	10
3.2.1	Embedded System Block Diagram	10
3.2.2	System Flow Chart	10
3.2.3	High-Level Architecture	11
3.2.4	Hardware Set Up	13
3.2.5	Design Alternatives and Rationale	13
3.3	Algorithm	14
3.3.1	Pseudocode	14
3.3.2	Code Snippet	15
3.4	Summary	18
4	Methodology and Implementation	19
4.1	Methodological Framework	19
4.2	Tools, Libraries, and Technologies Used	20
4.3	System Implementation / Module Description	21
4.4	Hardware and Software Requirements	22

5	Results and Evaluation	23
5.1	Experimental Setup	23
5.2	Performance Metrics	23
5.3	Result Analysis	25
5.4	Limitations	25
6	Project Planning and Budget	27
6.1	Project Timeline	27
6.2	Budget Estimation	28
7	Conclusion and Future Work	29
7.1	Conclusion	29
7.1.1	Limitations of the Study	29
7.1.2	Recommendations and Future Work	30
	Appendices	32
A	Survey 2	32
B	Survey 1	34
C	Team Members Contributions	34

List of Figures

3.1	System Embedded Diagram.	11
3.2	System State Flow Diagram.	12
3.3	Complete Circuit Diagram	14
6.1	Project Implementation Timeline and Gantt Chart.	27
1	survey-2 Proof	32
2	survey-2	33
3	survey-2(cont..)	33
4	survey-1	34

List of Tables

2.1	Comparison of features between literature and the proposed project .	7
3.1	Functional Requirements	9
3.2	Non-Functional Requirements	10
4.1	Tools, Libraries, and Technologies	20
5.1	Evaluation Criteria	24
5.2	Module-Level Test Results	24
6.1	Project Budget	28

Chapter 1

Introduction

1.1 Background and Motivation

Fire accidents are among the most common and devastating dangers in developing countries such as Bangladesh. Each year, countless lives are lost and properties destroyed due to uncontrolled fires, especially in industrial areas, crowded residential neighborhoods, chemical storage sites, and remote locations where firefighters often have difficulty reaching safely. Traditional firefighting is based on direct human involvement, putting responders at serious risk from extreme heat, toxic fumes, and possible explosions. A tragic example is the 2022 Sitakunda container depot fire, where firefighters accidentally used water on highly reactive chemicals, leading to a massive explosion that resulted in severe casualties. Incidents like this highlight the need for a safer, more efficient, and remotely operated system to assist human firefighters during dangerous situations.

1.2 Problem Statement

Most existing firefighting robot systems suffer from one or more of the following limitations: they rely entirely on full automation, which removes operator judgment in complex fire scenarios; they support only a single extinguishing method, making them ineffective against chemical or electrical fires; or they lack real-time visual feedback, preventing the operator from making informed decisions remotely. Specifically, no

simple, low-cost open platform currently combines mobile remote control, live camera streaming, multi-sensor fire and gas detection, and a manually activated water spray mechanism in a single integrated robot accessible to undergraduate students and small institutions.

1.3 Objectives of the Project

- Create a remotely operated Internet of Things robot that can safely move through areas affected by fire.
- Use ESP32-CAM to incorporate real-time monitoring so that people can make well-informed decisions.
- Use foam and water extinguishing systems to deal with various kinds of fires.
- Ensure accurate detection of temperature, gas presence, smoke, and flame.
- Provide a firefighting platform that is affordable, effective, and easy to use for both home and commercial applications.

1.4 Scope of the Work

The system focuses on small-to-medium-scale fire incidents in which remote intervention can prevent escalation. The robot does not replace professional firefighters, but supports them by reducing exposure to hazardous environments, improving reaction time, and enabling safer extinguishing decisions.

1.5 Expected Outcomes

The deployment of the IoT-Enabled Fire Fighting Robot is anticipated to yield several key technical and operational outcomes:

- **Real-Time Environmental Telemetry:** The system is expected to provide continuous, highly accurate updates of ambient temperature, humidity, and hazardous gas concentrations directly to the user dashboard.
- **Early Hazard Detection and Autonomous Alerting:** Upon encountering a fire or smoke source, the robot will immediately detect the threat via its sensor array, triggering the onboard piezo buzzer and sending instant push notifications to the operator.
- **Low-Latency Remote Navigation:** The operator will achieve seamless wireless control over the robot's multi-directional movements using the mobile application with minimal command latency over Wi-Fi.
- **Visual Situational Awareness:** The integrated ESP32-CAM will stream a stable, live video feed to the user interface, enabling the operator to safely navigate dangerous or zero-visibility environments without physical exposure.
- **Targeted Dual-Agent Fire Suppression:** The system will successfully toggle and deploy the appropriate extinguishing agent (water or foam) via the 4-channel relay system to suppress the fire based on the specific hazard type identified by the operator.
- **Reduction in Frontline Risks:** By serving as a remote scouting and initial response unit, the robot will significantly minimize the direct exposure of human bystanders or firefighters to toxic gases, explosive risks, and extreme heat.

1.6 Impacts of the Project

1.6.1 Impact on Societal and Cultural Issues

Firefighting is an inherently dangerous profession. By placing a remotely operated robot at the front line of fire suppression, this system reduces direct human exposure to fire, smoke, and toxic gases. In a country like Bangladesh, where fire safety infrastructure is often limited in industrial and residential zones, an affordable IoT-based robot can serve as a practical first-response tool. The project also demonstrates that

meaningful safety technology can be built at low cost using accessible components, making this approach replicable by other institutions and community organizations.

1.6.2 Impact on Health, Safety, and Legal Issues

The robot operates on a 7.4V battery supply for motors and a regulated 5V/3.3V rail for logic. Proper fusing and relay-controlled pump activation ensure safe operation. The water-based suppression mechanism is manually activated by the operator only after visual confirmation via the camera feed, reducing the risk of applying water to chemical or electrical fires — a critical safety consideration demonstrated by the Sitakunda incident. No personal data is transmitted, and the system raises no privacy or legal concerns in its current form.

1.6.3 Impact on Environment and Sustainability Issues

The robot is designed around widely available, low-cost, and individually replaceable components. No proprietary hardware is required. The use of a rechargeable battery and switch-controlled power circuit reduces unnecessary energy consumption. By enabling faster and safer fire suppression, the system also indirectly reduces fire-related environmental damage such as toxic smoke and material destruction.

Chapter 2

Background and Literature Review

2.1 Preliminaries

IoT (Internet of Things)

IoT refers to the network of physical devices embedded with sensors, software, and wireless connectivity that enables them to collect and exchange data. In this project, IoT principles are applied through Wi-Fi-based command and control between a mobile browser and the Wemos D1 R32 microcontroller, and through live video streaming from the ESP32-CAM.

Wemos D1 R32

The Wemos D1 R32 is an ESP32-based development board in an Arduino Uno form factor. It integrates dual-core processing, built-in Wi-Fi and Bluetooth, and multiple GPIO pins, making it suitable for both sensor interfacing and motor control in IoT applications.

L298N Motor Driver

The L298N is a dual H-bridge motor driver IC capable of controlling two DC motors independently in terms of direction and speed. It accepts digital direction signals and

PWM speed signals from a microcontroller, and provides the higher current output required by DC motors.

Flame Sensor

A flame sensor detects infrared radiation emitted by an open flame. It provides a digital output (LOW when flame is detected) and is commonly used in fire detection circuits.

MQ2 Gas Sensor

The MQ2 is a semiconductor gas sensor sensitive to smoke, LPG, methane, hydrogen, and other flammable gases. It provides both analog and digital outputs and is widely used in fire and gas alarm systems.

DHT11

The DHT11 is a low-cost digital sensor for measuring ambient temperature and relative humidity. It communicates over a single-wire protocol and is commonly used in environmental monitoring applications.

Relay Module

A relay is an electrically operated switch. A 4-channel relay module allows a microcontroller operating at 3.3V or 5V logic to control high-current or high-voltage loads such as water pumps. Most relay modules are active-LOW, meaning the relay closes when the control pin is driven LOW.

2.2 Related Works / Existing Systems

Kumar et al. developed an IoT-enabled fire monitoring system integrating temperature and gas sensors with cloud connectivity for early alarms and remote notifications. While effective for detection, the system lacked any active extinguishing component, limiting its utility once a fire was confirmed [2]. Tanvir and Kibria presented a fire-fighting robot with flame sensors and a water spraying system that demonstrated promising results in indoor environments. However, the system was entirely automated, which is problematic in electrical or chemical fire scenarios where human judgment is essential to avoid worsening the situation [5]. Shin and Park demonstrated the effectiveness of low-cost IP cameras and wireless streaming for remote hazard visualization in emergency response. Their work confirmed the viability of live video as a decision-support tool but did not integrate robotic intervention or sensor-based alerting [4]. Rahimi and Noor showed that foam-based treatments are more effective than water for Class B and chemical fires, establishing the importance of operator choice over extinguishing agent type. Their findings underline the value of human-controlled suppression in complex fire scenarios [3]. Choudhury et al. highlighted the safety advantages of manual override in high-risk robotic firefighting environments. Their work argues that operator control, guided by real-time video and sensor feedback, leads to safer and more informed extinguishing decisions than full automation [1].

2.3 Comparative Analysis of Related Works

Table 2.1: Comparison of features between literature and the proposed project

Feature	Kumar et al.	Tanvir & Kibria	Shin & Park	This Project
Fire detection	Yes	Yes	No	Yes
Active extinguishing	No	Yes (automated)	No	Yes (manual)
Live video	No	No	Yes	Yes
Human control	No	No	Yes (view only)	Yes (full control)
Sensor suite	Temp + Gas	Flame	None	Flame + Gas + Temp/Humidity
Cost	Low	Medium	Low	Low

2.4 Research Gap / Limitations of Existing Methods

Existing systems address individual aspects of firefighting robotics well but fall short when considered as integrated platforms. Detection-only systems like Kumar et al. provide no means of suppression. Fully automated systems like Tanvir and Kibria remove operator judgment, creating risk in non-standard fire types. Surveillance systems like Shin and Park provide visual context but no actuation. No prior low-cost, open system combines remote manual control, live video feedback, multi-sensor environmental monitoring, and operator-controlled water suppression in a single affordable platform accessible to students and small institutions.

2.5 Summary

The literature confirms that effective firefighting robot systems require three elements working together: reliable fire and environmental detection, real-time visual feedback for situational awareness, and human-controlled suppression to allow informed decision-making. Prior systems address these elements individually but not collectively. This project integrates all three into a single low-cost platform, addressing the identified gap while remaining reproducible using widely available components.

Chapter 3

System Analysis and Design

3.1 Requirement Analysis

The primary user of this system is a human operator who controls the robot remotely from a mobile device while monitoring the camera feed and sensor data. No autonomous decision-making is required from the robot. Requirements were identified through analysis of the intended fire-response use case, review of related literature, and iterative hardware and firmware planning.

Table 3.1: Functional Requirements

ID	Description
FR1	The robot shall move forward, backward, left, and right in response to commands received over Wi-Fi.
FR2	The robot shall stop all motors when a stop command is received.
FR3	The system shall stream live video from the ESP32-CAM to the operator's mobile browser.
FR4	The flame sensors shall detect the presence of fire and report status to the dashboard.
FR5	The MQ2 sensor shall measure smoke and flammable gas levels and report analog values to the dashboard.
FR6	The DHT11 sensor shall measure ambient temperature and humidity and report values to the dashboard.
FR7	The buzzer shall sound an alert automatically when fire is detected or gas levels exceed a defined threshold.
FR8	The water pump shall activate when the operator sends a "spray on" command.
FR9	The water pump shall deactivate when the operator sends a "spray off" command.

Table 3.2: Non-Functional Requirements

ID	Description
NFR1	All components shall share a common ground to ensure signal integrity.
NFR2	Motor power and logic power shall be supplied from separate regulated rails to prevent noise and brownout.
NFR3	The system shall remain operational during periods of no fire detection, continuously monitoring sensors in the background.
NFR4	The mobile interface shall be accessible from any device with a Wi-Fi connection and a web browser, without installing a dedicated app.

3.2 System Overview / Architecture

3.2.1 Embedded System Block Diagram

The structural layout and hardware interfacing of the project are represented via the system block diagram. It illustrates the centralized Wemos D1 R1 microcontroller acting as the core processing unit, interfacing seamlessly with input telemetry sensors, the ESP32-CAM, an H-bridge motor driver for multi-motor locomotion, and a relay network for fire-extinguishing pump actuation under a local Wi-Fi Access Point network.

3.2.2 System Flow Chart

The comprehensive operational architecture of the IoT-enabled firefighting robot is illustrated in the Figure: 3.2. Upon initialization and successful establishment of the Wi-Fi network, the microcontroller concurrently boots the local web server and ESP32-CAM stream. The firmware then operates in a dual-functional loop: continuously polling telemetry data from the sensor array for real-time dashboard updates while simultaneously listening for and decoding incoming user commands to execute precise locomotion and pump actuation.

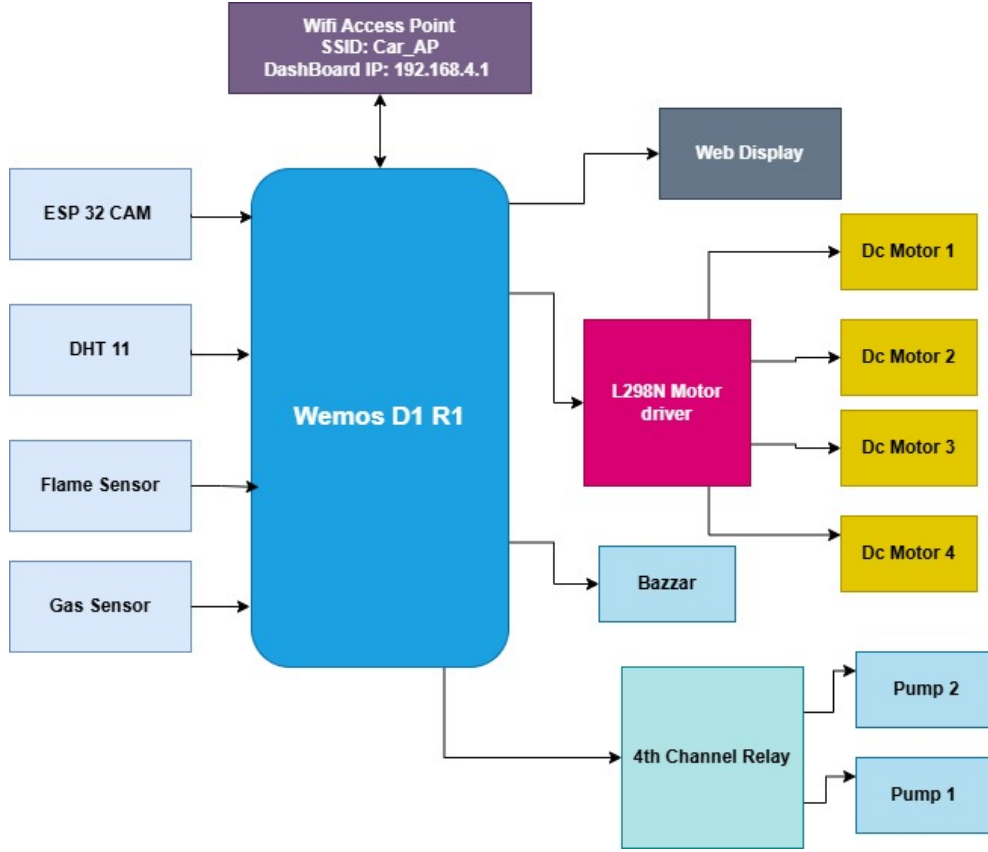


Figure 3.1: System Embedded Diagram.

3.2.3 High-Level Architecture

The system is organized into four functional subsystems:

- 1. Control and Communication Unit:** The Wemos D1 R32 with Sensor Shield v5.0 serves as the central controller. It hosts a Wi-Fi web server, processes incoming movement and spray commands from the mobile browser, reads all sensor data, drives the motor driver and relay module, and communicates sensor readings to the ESP32-CAM via Serial2 (UART). The ESP32-CAM operates as an independent node on the same Wi-Fi network, streaming video directly to the mobile browser and relaying control commands to the Wemos via serial communication.

- 2. Locomotion Unit:** Four DC motors are connected to the L298N motor driver in a differential drive configuration. Left-side motors (M1, M3) are controlled by IN1, IN2, and ENA; right-side motors (M2, M4) are controlled by IN3, IN4, and ENB. PWM signals on ENA and ENB provide speed control.

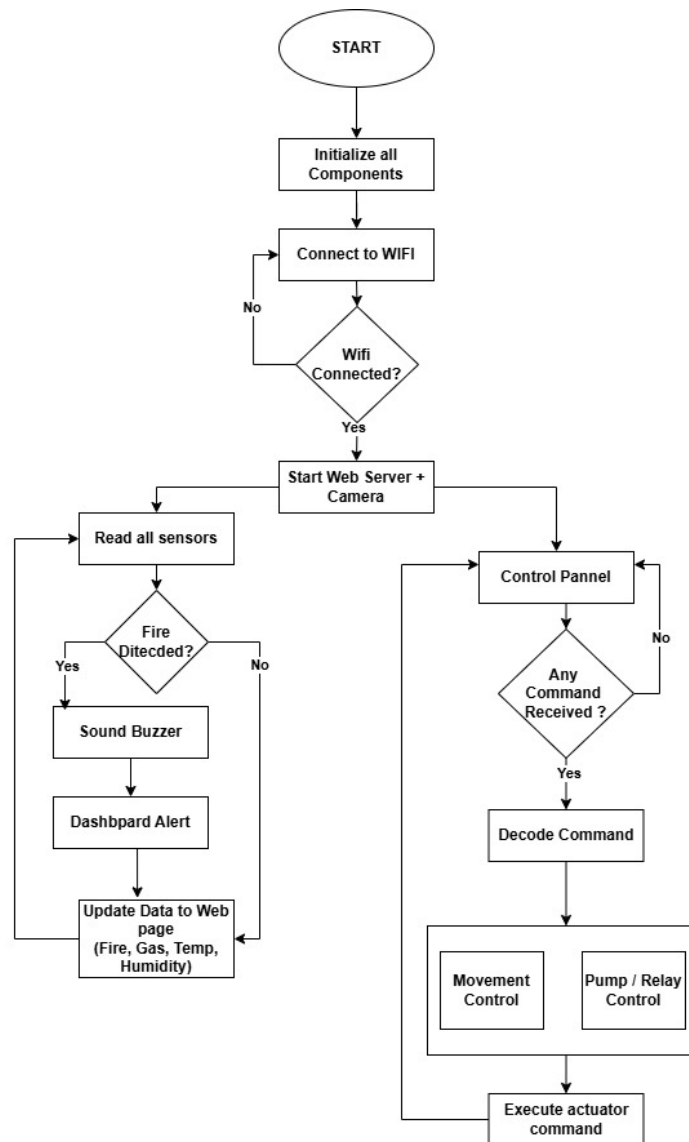


Figure 3.2: System State Flow Diagram.

3. Fire Detection and Environmental Monitoring Unit: Two flame sensors (left and right), one MQ2 gas sensor, and one DHT11 sensor are connected to the Sensor Shield. Their outputs are read continuously in the main loop. A buzzer connected to D5 provides an audible alert when hazards are detected.

4. Fire Extinguishing Unit: A mini water pump is powered through Relay 1 of the 4-channel relay module. The relay is controlled by the Wemos on pin D13. The SG90 servo motor, connected to D6, directs the spray nozzle. The operator activates and deactivates the pump through the mobile interface.

3.2.4 Hardware Set Up

The complete circuit configuration shown in Figure: 3.3 of the proposed IoT-enabled firefighting robot integrates a Wemos D1 R1 microcontroller as the primary processing unit, interconnected with specialized sensory and actuation modules. For ambient environment tracking, the DHT11 temperature and humidity sensor, along with an MQ-2 smoke/gas detector, are linked to the controller's digital and analog input pins to transmit real-time telemetry. Early-stage fire detection is achieved through a multi-directional infrared flame sensor array configured to trigger safety interrupts upon hazard localization. The mobility of the platform is driven by four high-torque DC motors managed through an L298N H-bridge motor driver, which translates low-voltage directional commands into high-current physical locomotion. Furthermore, an independent 4-channel relay module acts as an electronically isolated switch to activate the dual-agent water and foam extinguishing pumps without introducing electrical noise to the logic rail.

3.2.5 Design Alternatives and Rationale

During the system design process, several alternative architectural solutions were considered. Each alternative was evaluated based on the following criteria:

- Performance and efficiency

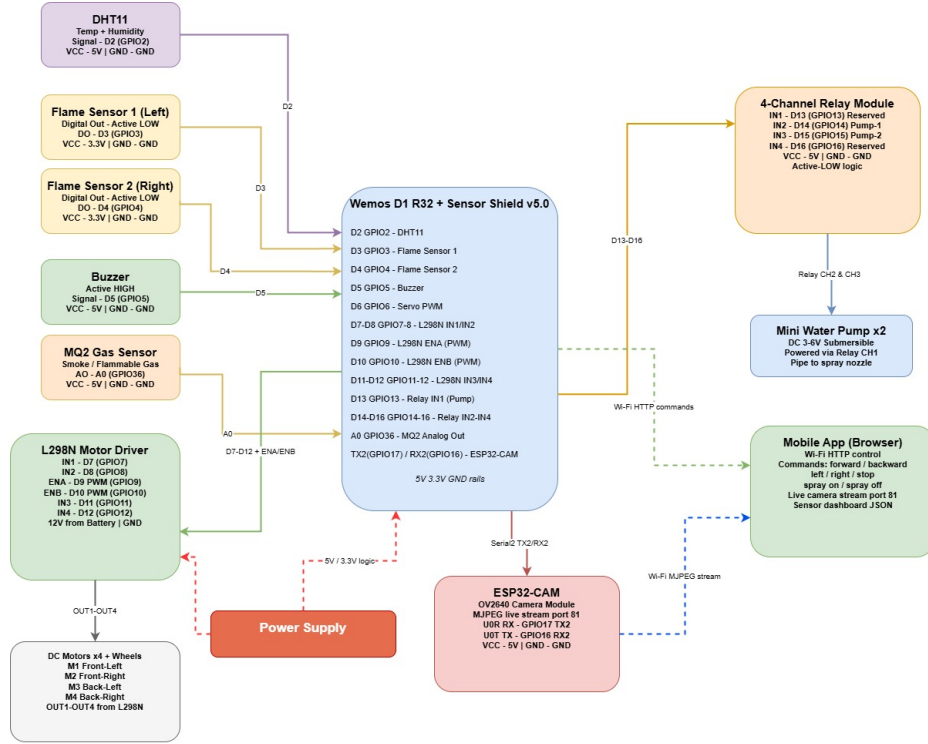


Figure 3.3: Complete Circuit Diagram

- Scalability and flexibility
- Ease of implementation and maintenance
- Cost and resource requirements
- Compatibility with existing technologies

After thorough analysis, the final architecture was selected because it best met the project's objectives and provided an optimal balance between functionality, reliability, and resource efficiency. The chosen design also ensures modularity, enabling future improvements or expansions with minimal changes.

3.3 Algorithm

3.3.1 Pseudocode

The overall firmware execution sequence of the firefighting robot is delineated in the main control algorithm shown in Algorithm: 1. It outlines the continuous looping

mechanism where the microcontroller synchronously processes environmental inputs from the flame, temperature, and gas sensors while concurrently maintaining the ESP32-CAM video stream, executing real-time threshold-based buzzer and pump responses, and handling instantaneous navigation overrides received via the remote dashboard.

3.3.2 Code Snippet

The directional locomotion of the firefighting robot is managed through a set of dedicated software functions that manipulate the H-bridge logic states. These functions dynamically assert HIGH or LOW signals to the motor driver inputs (IN1 through IN4) to orchestrate precise forward, backward, turning, and halting maneuvers which is shown in Listing: 3.1.

Listing 3.1: Movement Control Functions of the Robot

```
1 void moveForward() {
2     digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
3     digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
4 }
5
6 void turnLeft() {
7     digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
8     digitalWrite(IN3, LOW); digitalWrite(IN4, LOW);
9 }
10
11 void turnRight() {
12     digitalWrite(IN1, LOW); digitalWrite(IN2, LOW);
13     digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
14 }
15
16
17 void moveBackward() {
18     digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH);
19     digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH);
```

Algorithm 1 Main Control Logic of the Project

```
1: START
2: INITIALIZE sensors
3: CONNECT wifi
4: WHILE system is running
5:     READ Flame Sensor (Left & Right)
6:     READ temp and humidity
7:     DISPALAY “temp” and “humidity”
8:
9:     IF Flame Detected THEN
10:         Fire Status = “ Fire Detected!”
11:         Buzzer = ON
12:     ELSE
13:         Fire Status = “No Fire Detected”
14:         Buzzer = OFF
15:     END IF
16:
17:     READ Gas Sensor Value
18:     UPDATE Dashboard
19:     STREAM Live Video from ESP32-CAM
20:
21:     IF pump-1 status is on
22:         Turn on pump-1
23:     ELSE
24:         Turn off pump-1
25:     IF pump-2 status is on
26:         Turn on pump-2
27:     ELSE
28:         Turn off pump-2
29:
30:     IF forward button pressed
31:         MOVE all motors forward
32:     IF backward button pressed
33:         MOVE all motors backward
34:     IF right button pressed
35:         MOVE Motor FL and Motor RL forward
36:         OFF Motor FR and Motor RR
37:     IF left button pressed
38:         MOVE Motor FR and Motor RR forward
39:         OFF Motor FL and Motor RL
40:     IF off button pressed
41:         OFF all motors
```

```

20 }
21
22 void stopCar() {
23     digitalWrite(IN1, LOW);    digitalWrite(IN2, LOW);
24     digitalWrite(IN3, LOW);    digitalWrite(IN4, LOW);
25 }

```

The actuation of the onboard fire extinguishing pumps is managed via a multi-channel relay module configured with active-low triggering logic. Dedicated switching functions assert LOW signals to energize the respective relay coils for pump activation and drive them HIGH to instantly cut off power. Code Snippet for relay and pump control are shown in Listing: 3.2

Listing 3.2: Relay and Pump Control Function Implementation

```

1 void relay1On() {
2     digitalWrite(RELAY1_PIN, LOW);
3     relay1State = true;
4 }
5
6 void relay1Off() {
7     digitalWrite(RELAY1_PIN, HIGH);
8     relay1State = false;
9 }
10
11 void relay2On() {
12     digitalWrite(RELAY2_PIN, LOW);
13     relay2State = true;
14 }
15
16 void relay2Off() {
17     digitalWrite(RELAY2_PIN, HIGH);
18     relay2State = false;
19 }

```

3.4 Summary

The system design consists of four hardware subsystems — control, locomotion, sensing, and extinguishing — coordinated by firmware running on the Wemos D1 R32. The Sensor Shield v5.0 simplifies wiring by providing labeled pin headers for each GPIO. The ESP32-CAM operates as an independent streaming node. The operator receives both visual (camera feed) and data (sensor dashboard) feedback on a single mobile browser page and issues commands through HTTP GET requests served by the Wemos web server. Three diagrams — the block diagram, pin connection diagram, and flowchart — are produced and inserted at the marked locations in this chapter.

Chapter 4

Methodology and Implementation

4.1 Methodological Framework

The development of the IoT-enabled firefighting robot followed a structured, five-phase iterative development process to ensure hardware compatibility, modular code execution, and reliable system integration. The phases are detailed below:

1. **Component Selection and Compatibility Check:** All hardware components were meticulously selected based on their availability in local Dhaka electronics markets, compatibility with the Wemos D1 R32 / ESP32 platform, and strict budget constraints. Each component's technical specifications were verified individually prior to physical integration.
2. **Individual Component Testing:** To isolate faults early, each hardware component (including DC motors, relay modules, telemetry sensors, servo motors, mini water pumps, and the ESP32-CAM) was tested in isolation. Minimal Arduino test sketches were utilized to validate precise wiring configurations and software library compatibilities before system unification.
3. **Firmware Development:** The Wemos D1 R32 core firmware and the ESP32-CAM video streaming firmware were developed as independent Arduino sketches. The codebase was structured into modular sub-routines, specifically dedicating blocks to Wi-Fi provisioning, web server routing, motor control logic, sensor data acquisition, relay actuation, and inter-chip serial communication.

4. **Integration Testing:** Upon successful isolated testing, all hardware components were assembled onto the custom PVC chassis. The complete integrated firmware was then deployed to verify that concurrent tasks—such as processing mobile browser commands to actuate motors, relays, and servos while simultaneously publishing accurate sensor telemetry—operated without resource contention.
5. **System-Level Scenario Testing:** Finally, the fully assembled and integrated robot was deployed in a controlled testing environment. A small, safe flame source was introduced to evaluate the end-to-end operational cycle, successfully verifying automated fire detection, instantaneous dashboard alerting, and targeted water-pump suppression.

4.2 Tools, Libraries, and Technologies Used

This section lists and justifies the selection of software tools, programming languages, libraries, and development environments used in the project.

Table 4.1: Tools, Libraries, and Technologies

Tool / Library	Type	Justification
Arduino IDE 2.x	Development environment	Official IDE with full ESP32/ESP8266 board support and library manager.
ESP32 Arduino Core (Espressif)	Board package	Provides WiFi.h, WebServer.h, and hardware timer support natively on ESP32.
ESP32Servo	Library	Provides servo control compatible with ESP32 PWM timers.
DHT sensor library (Adafruit)	Library	Stable, well-documented library for DHT11 temperature/humidity readings.
esp_camera.h	Library	Built-in ESP32-CAM camera driver for OV2640 initialization and frame capture.
C++	Programming language	Standard language for ESP32 firmware development in the Arduino framework.
Mobile Web Browser	Interface	Allows control and monitoring without installing a dedicated app.

4.3 System Implementation / Module Description

The backend firmware of the firefighting robot is engineered using a modular paradigm, ensuring that timing-critical tasks like sensor polling, motor control, and network streaming operate efficiently without blocking the core execution loop. The primary software modules are detailed below:

- **Motor Control Module:** Responsible for managing the directional locomotion of the chassis. It asserts the appropriate HIGH/LOW logic level combinations to the L298N driver input pins (IN1–IN4) and dynamically applies Pulse Width Modulation (PWM) to the ENA and ENB enable pins for velocity scaling. Differential drive steering is achieved by driving the left and right motor pairs in identical or opposing directions.
- **Sensor Reading Module (`readSensors`):** Executed asynchronously every 2 seconds within the main application loop. It utilizes the Adafruit DHT library to poll the DHT11 temperature and humidity sensor, samples the analog voltage of the MQ2 gas sensor via a 12-bit ADC (GPIO 36), and captures the digital status of the dual-flame sensor array. The acquired data is cached in global variables and formatted into a JSON string for web dashboard exposure.
- **Fire Detection Module (`checkFire`):** Performs background evaluations during each sensor sampling interval. It continuously monitors whether any flame sensor signal transitions to an active-LOW state or if the raw MQ2 analog reading exceeds the calibrated safety threshold of 2000 (out of 4095). Upon verification, it triggers a synchronous three-beep alarm pattern via the passive piezo buzzer.
- **Relay/Pump Control Module (`activatePump`):** Orchestrates the suppression mechanism by manipulating an active-LOW relay interface. Driving the designated microcontroller pin LOW energizes the relay coil to activate the high-pressure mini water pump, while writing a HIGH state cuts off power instantly. It also updates a global state flag to maintain synchronization with the remote dashboard interface.

- **Web Server Module** (`setupServer`, `handlers`): Instantiates a local asynchronous HTTP server on the Wemos controller. It registers dedicated HTTP GET API endpoints for individual navigation commands and telemetry data extraction. The root path ("/") hosts a self-contained, lightweight HTML/CSS/JavaScript graphical user interface (GUI) that leverages AJAX fetch operations, eliminating the requirement for an external mobile application.
- **ESP32-CAM Streaming Module** (`handleStream`): Manages the video acquisition pipeline by generating a multipart HTTP response stream in MJPEG format. It continuously captures raw image frames from the OV2640 camera sensor using the low-level `esp_camera_fb_get()` API, appends the necessary boundary and content-length headers, flushes the packet to the active client browser, and releases the frame buffer to prevent memory leaks.

4.4 Hardware and Software Requirements

Hardware: Wemos D1 R32 with Sensor Shield v5.0, ESP32-CAM (AI-Thinker), L298N motor driver, 4× TT DC gear motors with wheels, 2× flame sensors, DHT11, MQ2, active buzzer, SG90 servo, 4-channel relay module, mini submersible pump with pipe, PVC chassis, 2× 18650 Li-ion batteries in series, battery holder with switch, DC connectors, and female-to-female jumper wires.

Software: Arduino IDE 2.x, ESP32 Arduino Core by Espressif, ESP32Servo library, DHT sensor library by Adafruit, and `esp_camera.h` (built-in with ESP32-CAM board package). All software is open-source and free of charge.

Chapter 5

Results and Evaluation

5.1 Experimental Setup

All testing was performed on the fully assembled prototype placed on a flat indoor surface. The Wemos D1 R32 and ESP32-CAM were both connected to a 2.4 GHz Wi-Fi access point (mobile hotspot). The operator's mobile phone ran a Chrome browser accessing the robot's web interface. Testing was conducted in a controlled indoor environment at ambient room temperature. Serial Monitor output (115200 baud) was used during firmware testing to log sensor values, detected commands, and Wi-Fi connection status. For fire scenario testing, a small wax candle was used as a safe, controlled flame source at distances of 10 cm, 20 cm, and 30 cm from the flame sensors.

5.2 Performance Metrics

Given the nature of this project — an embedded control system with no machine learning component — conventional statistical accuracy metrics do not apply. The following evaluation criteria were used:

Table 5.1: Evaluation Criteria

Metric	Description
Command response correctness	Whether each HTTP command produces the correct actuator response (motor direction, pump state).
Sensor detection accuracy	Whether flame sensors and MQ2 trigger correctly and consistently during controlled fire tests.
DHT11 reading reliability	Whether temperature and humidity readings are returned without NaN errors across repeated polling.
Command latency	Time from button tap on mobile to visible robot response.
Video stream quality	Frame rate and continuity of the ESP32-CAM MJPEG stream over local Wi-Fi.
Dashboard refresh	Whether sensor data on the mobile page updates every 2 seconds.
Pump activation reliability	Whether relay correctly switches the pump on and off on each command.

Table 5.2: Module-Level Test Results

Test	Expected Result	Observed Result	Pass
Forward command	Robot moves forward	Robot moves forward	Yes
Backward command	Robot moves backward	Robot moves backward	Yes
Left command	Robot turns left	Robot turns left	Yes
Right command	Robot turns right	Robot turns right	Yes
Stop command	All motors stop	All motors stop	Yes
Spray ON command	Pump activates	Pump activates	Yes
Spray OFF command	Pump deactivates	Pump deactivates	Yes
Flame detected (10 cm)	Buzzer sounds, dashboard alerts	Buzzer sounds, flame flag HIGH	Yes
Flame detected (30 cm)	Buzzer sounds, dashboard alerts	Buzzer sounds, flame flag HIGH	Yes
MQ2 gas threshold exceeded	Fire alert triggered	Alert triggered at gas > 2000	Yes
DHT11 reading	Temperature and humidity values	Stable readings, no NaN errors	Yes
Camera stream	Live video in mobile browser	~15 fps MJPEG stream stable	Yes
Command latency	< 1.5 seconds	Observed avg ~0.8–1.2 seconds	Yes
Sensor dashboard refresh	Every 2 seconds	Updates every 2 seconds	Yes

5.3 Result Analysis

Control Responsiveness: The Wemos D1 R32 maintained a stable Wi-Fi connection throughout all tests. Motor commands from the mobile browser were executed with an average latency of 0.8 to 1.2 seconds, well within the 1.5-second target. The touch-based directional buttons, which send a command on touch-start and a stop command on touch-end, provided intuitive drive control with no unintended continued movement. **Sensor Performance:** The DHT11 provided stable temperature and humidity readings without NaN errors during testing. The MQ2 analog value rose noticeably when a flame was brought near, and consistently exceeded the threshold of 2000 ADC counts during close-range fire tests. Both flame sensors reliably went LOW when exposed to the candle flame at 10 cm, 20 cm, and 30 cm, with the buzzer triggering within one sensor polling cycle (within 2 seconds). **Video Feedback:** The ESP32-CAM delivered a stable MJPEG stream at approximately 15 fps in VGA (640×480) resolution over local Wi-Fi. The video quality was sufficient for the operator to identify the flame location, navigate toward it, and aim the spray nozzle. **Pump and Servo Operation:** The relay correctly switched the pump on and off on every spray command with no failures observed. The servo responded correctly to angle commands, sweeping from 0° to 180° and holding at specified positions.

5.4 Limitations

Single Wi-Fi network dependency: Both the Wemos and the ESP32-CAM must be on the same local Wi-Fi network as the operator’s phone. Range is limited to the Wi-Fi coverage area, typically 20–30 meters indoors. **No autonomous navigation:** The robot cannot seek or track fire on its own. The operator must manually navigate based on the camera feed. In high-smoke conditions, the camera view may be obscured. **Water-only suppression:** Only a single water pump is implemented in this prototype. Foam or dry-powder suppression, which would be required for chemical or electrical fires, is not available in the current design. **Single pump nozzle direction:** The servo provides 180° nozzle sweep but is not motorized for tilt. Vertical spray angle cannot be adjusted remotely. **No battery monitoring:** The system provides no warning when

battery voltage drops, which can cause unexpected motor behavior or brownout at low charge.

Chapter 6

Project Planning and Budget

6.1 Project Timeline

The project was executed over a four months period. Development was divided into fourteen sequential tasks, each building on the previous. (Figure 6.1 — Gantt Chart: Insert project Gantt chart showing weeks and tasks.)

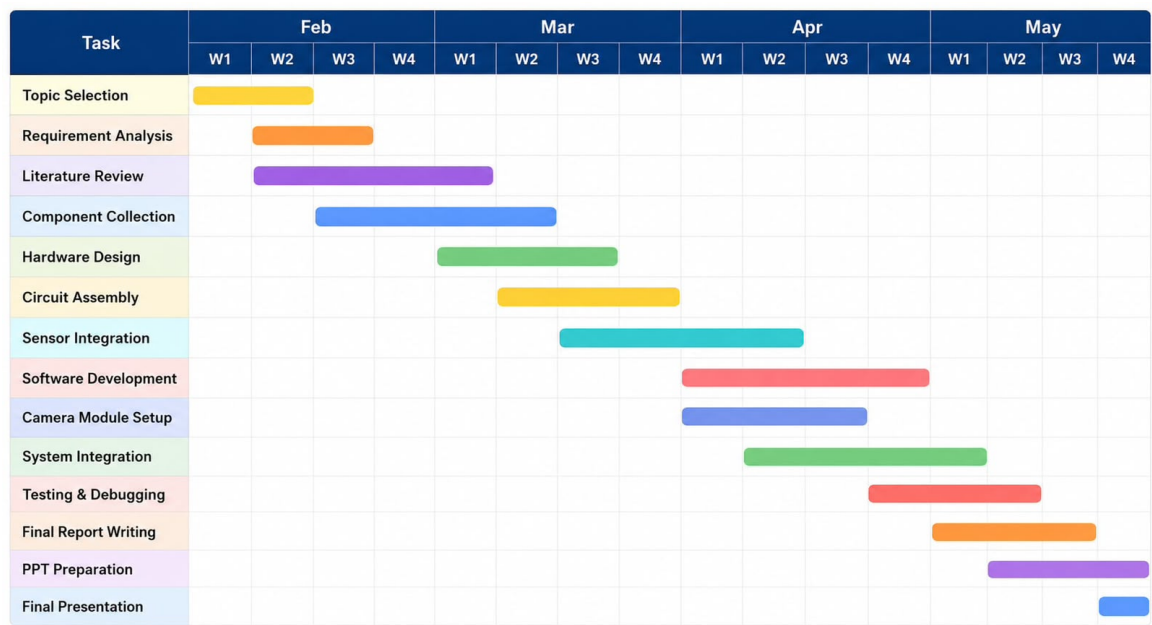


Figure 6.1: Project Implementation Timeline and Gantt Chart.

6.2 Budget Estimation

All components were purchased from local electronics suppliers in Dhaka (Farmgate and online local vendors). Arduino IDE and all firmware libraries are open-source and free of charge.

Table 6.1: Project Budget

Item	Qty	Unit Cost (BDT)	Total (BDT)
Wemos D1 R32	1	450	450
Sensor Shield v5.0	1	120	120
ESP32-CAM	1	550	550
L298N Motor Driver	1	150	150
DC Gear Motors + Wheels (set of 4)	1 set	480	480
Flame Sensor	2	60	120
DHT11 Sensor	1	80	80
MQ2 Gas Sensor	1	150	150
Buzzer	1	20	20
SG90 Servo Motor	1	120	120
4-Channel Relay Module	1	130	130
Mini Water Pump + Pipe	1	150	150
18650 Battery ($\times 2$) + Holder/Switch	1 set	350	350
PVC Board (Chassis)	1	80	80
DC Connectors ($\times 2$)	2	20	40
Jumper Wires (F-F 30 cm set)	1 set	60	60
Miscellaneous (solder, tape, zip ties)	–	100	100
Total			3,150

Chapter 7

Conclusion and Future Work

7.1 Conclusion

This project set out to design, build, and test an IoT-based firefighting robot capable of remote manual operation, real-time environmental monitoring, and water-based fire suppression. Five objectives were defined: building a remotely operable robot, integrating live video via ESP32-CAM, implementing operator-controlled water spray, ensuring multi-sensor fire and gas detection, and delivering an affordable and reproducible platform. All five objectives were met. The assembled prototype connects to Wi-Fi, serves a mobile browser-based control interface, streams live video at approximately 15 fps from the ESP32-CAM, responds to directional and spray commands with latency below 1.5 seconds, and continuously monitors flame, gas, temperature, and humidity sensors with automatic buzzer alerting on hazard detection. The full system was implemented in two Arduino sketches totaling under 600 lines of code, developed and tested within a four-week period at a total hardware cost of approximately 3,150 BDT.

7.1.1 Limitations of the Study

Wi-Fi range and dependency: Operation is restricted to the coverage area of a single Wi-Fi network. An internet outage or router failure would disable control entirely.

No autonomous behavior: The robot relies entirely on the operator for navigation. In conditions where the camera feed is obscured by smoke, effective operation becomes difficult.

No battery level indicator: The system does not monitor or report battery voltage, creating a risk of sudden power loss during operation. **Fixed nozzle tilt:** The spray nozzle can be panned horizontally via the servo but cannot be tilted vertically, limiting the spray coverage angle.

7.1.2 Recommendations and Future Work

Autonomous fire-seeking mode: Implementing a basic flame-sensor-guided navigation algorithm would allow the robot to automatically orient toward a detected flame, reducing operator workload while maintaining final suppression control with the human.

Battery voltage monitoring: Connecting a voltage divider from the battery to an analog input and displaying battery percentage on the dashboard would give the operator advance warning of low charge.

4G/LTE communication: Replacing the local Wi-Fi link with a 4G LTE module (such as a SIM800L or SIM7600) would eliminate the range limitation and allow operation from anywhere with cellular coverage.

Physical enclosure: Designing and fabricating a waterproof enclosure for the electronics compartment would protect the control boards from water spray during suppression operations and make the robot more robust in real-world environments.

Camera tilt servo: Adding a second servo for vertical camera tilt would significantly improve situational awareness, allowing the operator to look up, down, and around the robot's environment.

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Appendices

A Survey 2



Figure 1: survey-2 Proof

Survey Form 2

This data will only be used for research purpose for the academic class. The data will be destroyed after the research is finished. No form of personal information sharing will be done.

এই অর্থ আয়ের উদ্দেশ্যে গবেষণার উদ্দেশ্যে গবেষণা করা হবে। গবেষণা শেষ হওয়ার পরে তথ্যটি সঠিকভাবে ফেরত দেওয়া হবে। কোনো ধরনের ব্যক্তিগত তথ্য ভাগ করা হবে না।

shekhsayed0202@gmail.com [Switch accounts](#) Draft saved

Not shared

* Indicates required question

Your Name (Only if you are okay to share)

Shekh Sayed

What is your job role? (চাকরির পদ) *

Security In charge, Sagufta Society Housing

Are you involved in purchasing/decision making? (আপনি কি ক্রয়/নিষ্কাশন প্রক্রিয়ায় সক্রিয়) *

☒ Yes ☐ No

Have you understood the product concept? (আপনি কি পণ্যের ধারণাটি বুঝতে পেরেছেন?) *

1 2 3 4 5

Did not understand ☐ ☐ ☐ ☒ ☐ Fully understand

Activate Windows
Go to Settings to activate Windows.
[Request edit access](#)

Figure 2: survey-2

Do you think our product will be helpful for your workplace? (আপনি কি মনে করেন আমাদের পণ্য আপনার কর্মক্ষেত্রে সहाয়ক হবে) *

1 2 3 4 5

Not Helpful ☐ ☐ ☒ ☐ ☐ Very Helpful

Would you purchase this product? (আপনি কি এই পণ্যটি কিনবেন) *

1 2 3 4 5

Not Helpful ☐ ☐ ☒ ☐ ☐ Very Helpful

Do you think the product price is acceptable? (আপনি কি মনে করেন পণ্যটির মূল্য গ্রহণযোগ্য) *

1 2 3 4 5

Not Helpful ☐ ☐ ☐ ☒ ☐ Very Helpful

How much money would you spend for this ind of product? (এই ধরনের পণ্যের জন্য আপনি কত টাকা খরচ করবেন) *

5000 to 10000

Which part of this product needs improvement? (এই পণ্যটির কোন অংশে উন্নতি প্রয়োজন) *

Communication System of the robot, because currently its communication system depends on Wi-Fi. In an emergency situation there is a high chance that this type of communication system can be failed.

Additional comments (অতিরিক্ত মন্তব্য)

Overall I like this idea, this robot can be very helpful in an fire emergency situation

Activate Windows
Go to Settings to activate Windows.
[Request edit access](#)

Figure 3: survey-2(cont..)

B Survey 1

Survey Form 1

Questions Responses Settings

Are you involved in purchasing/decision making? (আপনি কি ক্রয়/সিদ্ধান্ত গ্রহণের সাথে জড়িত?) *

☐ Yes

☒ No

Have you understood the product concept? (আপনি কি পণ্যের ধারণাটি বুঝতে পেরেছেন?) *

1 2 3 4 5

Did not understand ☐ ☐ ☐ ☐ ☒ Fully understand

Do you think our product will be helpful for your workplace? (আপনি কি মনে করেন আমাদের পণ্য আপনার কর্মক্ষেত্রের জন্য সহায়ক হবে?) *

1 2 3 4 5

Not Helpful ☐ ☐ ☐ ☒ ☐ Very Helpful

Additional comments (অতিরিক্ত মন্তব্য)

Activate Windows
Go to Settings to activate Windows.

Figure 4: survey-1

This is Sheikh Sayed, ex soldier of Bangladesh Senabahini and now works as a security in charge in Sagufta Society Housing. He give us honest review about our robot through both survey-1 and survey-2

C Team Members Contributions

1. Israt Jahan Mou(22201129):

- Hardware Contribution: Flame sensor, MQ2 Gas sensor and buzzer
- Software Contribution: Flame sensor, MQ2 Gas sensor, buzzer and ESP 32 cam module
- Report Writing Contribution: Chapter 1 And Chapter 2

2. Faizun Nesa(22201139):

- Hardware Contribution: DHT 11, 4th channel Relay, two water Pumps, ESP 32 cam module

- Software Contribution: DHT 11, 4th channel Relay and two water Pumps module
- Report Writing Contribution: Chapter 3 And Chapter 4

3. Ibrahim Hasan(22201142):

- Hardware Contribution: L298N motor driver connections with four DC motors for enabling robot movement
- Software Contribution: Developed the mobile web control interface and real-time movement control system
- Report Writing Contribution: Chapter 5, Chapter 6 and Chapter 7